

Bot 2103 Short Questions

1. Enumerate the fundamental principles of chromosomes theory of inheritance.

1. Chromosomes contain the genetic material that is transmitted from parent to offspring and from cell to cell.
2. Chromosomes are replicated and passed along, generation after generation, from parent to offspring. They are also passed from cell to cell during the development of a multicellular organism. Each type of chromosome retains its individuality during cell division and gamete formation.
3. The nuclei of most eukaryotic cells contain chromosomes that are found in homologous pairs—they are diploid. One member of each pair is inherited from the mother, the other from the father. At meiosis, one of the two members of each pair segregates into one daughter nucleus, and the homolog segregates into the other daughter nucleus. Gametes contain one set of chromosomes—they are haploid.
4. During the formation of haploid cells, different types of (nonhomologous) chromosomes segregate independently of each other.
5. Each parent contributes one set of chromosomes to its offspring. The maternal and paternal sets of homologous chromosomes are functionally equivalent; each set carries a full complement of genes.

2. Write short note on trisomy in plants.

- In general, the effects of trisomy ($2n + 1$) are similar to those of monosomy.
- However, the addition of an extra chromosome produces somewhat more viable individuals in both animal and plant species than does the loss of a chromosome.
- In animals, this is often true, provided that the chromosome involved is relatively small.
- However, the addition of a large autosome to the diploid complement in both *Drosophila* and humans has severe effects and is usually lethal during development.

3. Classify the two-point test crossing over between two genes.

- The percentage of crossing over between two linked genes is calculated by test crosses in which a F_1 dihybrid is crossed with a double recessive parent.
- In such crosses, crossing overs occur at two points, hence it is called **two-point test cross**.
- For example, a dihybrid having the genotype AC/ac is test crossed with a double recessive parent ac/ac and test cross produces 74% parental combinations and 26% cross over types.
- Then, the distance between the loci A and C is estimated to be 26 centimorgans.
- If the distance between the two loci is large in two-point test cross, double cross over types are undetected and it would appear as parental types.
- Hence, the true map distance (crossover percentage) is underestimated.

Bot 2013 Long Question

1. Discuss the pattern of X-linkage in human with suitable diagram. Art: 3.7.2 Fig. 3.10
 - In humans, many genes and the respective traits controlled by them are recognized as being linked to the X chromosome.
 - These X-linked traits can be easily identified in a pedigree because of the crisscross pattern of inheritance. A pedigree for one form of human color blindness is shown in Figure 3.10.
 - The mother in generation I passes the trait to all her sons but to none of her daughters.
 - If the offspring in generation II marry normal individuals, the color-blind sons will produce all normal male and female offspring (III-1, 2, and 3); the normal-visioned daughters will produce normal-visioned female offspring (III-4, 6, and 7), as well as color-blind (III-8) and normal-visioned (III-5) male offspring.
 - The way in which X-linked genes are transmitted causes unusual circumstances associated with recessive X-linked disorders, in comparison to recessive autosomal disorders.
 - For example, if an X-linked disorder debilitates or is lethal to the affected individual prior to reproductive maturation, the disorder occurs exclusively in males.
 - This is so because the only sources of the lethal allele in the population are in heterozygous females who are "carriers" and do not express the disorder.
 - They pass the allele to one-half of their sons, who develop the disorder because they are hemizygous but rarely, if ever, reproduce.
 - Heterozygous females also pass the allele to one-half of their daughters, who become carriers but do not develop the disorder.
 - An example of such an X-linked disorder is Duchenne muscular dystrophy. The disease has an onset prior to age 6 and is often lethal around age 20. It normally occurs only in males.